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Graduate Research Symposium

Assessment of Atmospheric Plume Source Inversion using Sparse Reconstruction and Gaussian Dispersion

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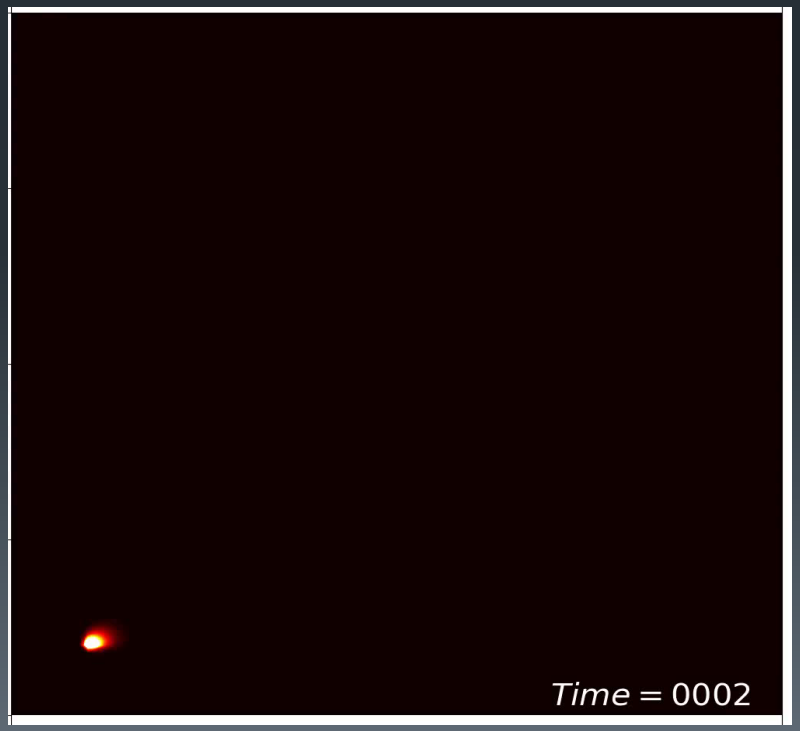
Outline

- Introduction to atmospheric dispersion and it's modeling.
- Forward models:
  - Gaussian Dispersion (GD) model.
  - 3D unsteady advection diffusion scalar Transport Equation (TE).
- Study on the errors associated with the GD Model.
- Inverse Models
  - POD based sparse reconstruction using TE.
  - GD model based inversion.
- Modeling the error (ongoing research).

Introduction

- Atmospheric dispersion modeling is the mathematical description of contaminant transport.
- To keep track of the contaminants in ambient atmosphere and minimize toxic effects efficiently.
- Estimating leak rate and source location of potentially hazardous gas emissions.

Video: Full field LES simulation of atmospheric dispersion originated from a point source



Major Goals

- Building a forward model
  - 3 ways:
    - Gaussian Dispersion (GD) model – Idealized
    - LES Model – Costly
    - Mean advected Transport equation – Middle ground
- Inverse problem
  - Sparse reconstruction
  - GD model based reconstruction

Forward Models

- Input:
  - Advective velocity field
  - Leak rate
  - Leak source location
  - Diffusion parameter
- Output:
  - Concentration field

Gaussian Dispersion (GD) Model

- Model equation:

$$C(x,y,z) = \frac{Q}{2\pi\sigma_y\sigma_z u} e^{-\frac{y^2}{2\sigma_y{}^2}} [e^{-\frac{(z-h)^2}{2\sigma_z{}^2}} + e^{-\frac{(z+h)^2}{2\sigma_z{}^2}}]$$

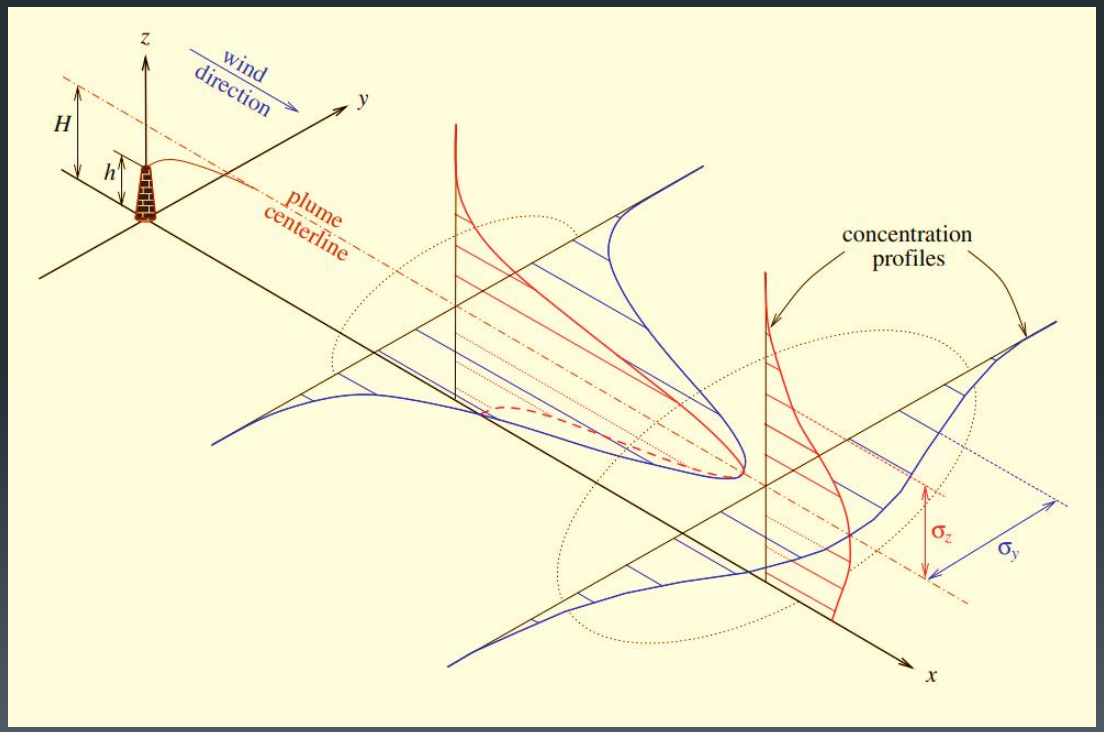


Figure : Gaussian shape of contaminant plume emitted from a point source (J. M. Stockie, SIAM Review, 2012)

Assumptions in GD Model

- Steady state solution.
- Eddie difusivities are function of streamwise distance only.
- Constant wind velocity.
- Constant emission rate from a single point source.
- Advection is more dominant over diffusion in x direction.
- z = 0 at the ground.
- Contaminate does not penetrate the ground.

Transport Equation (TE)

- Governing equation:

$$\frac{\partial C}{\partial t} + u.\nabla C = q + k\nabla^2 C$$

- Here the source term *q* is expressed as following where *Q* is the constant emission rate and δ is Dirac delta function:

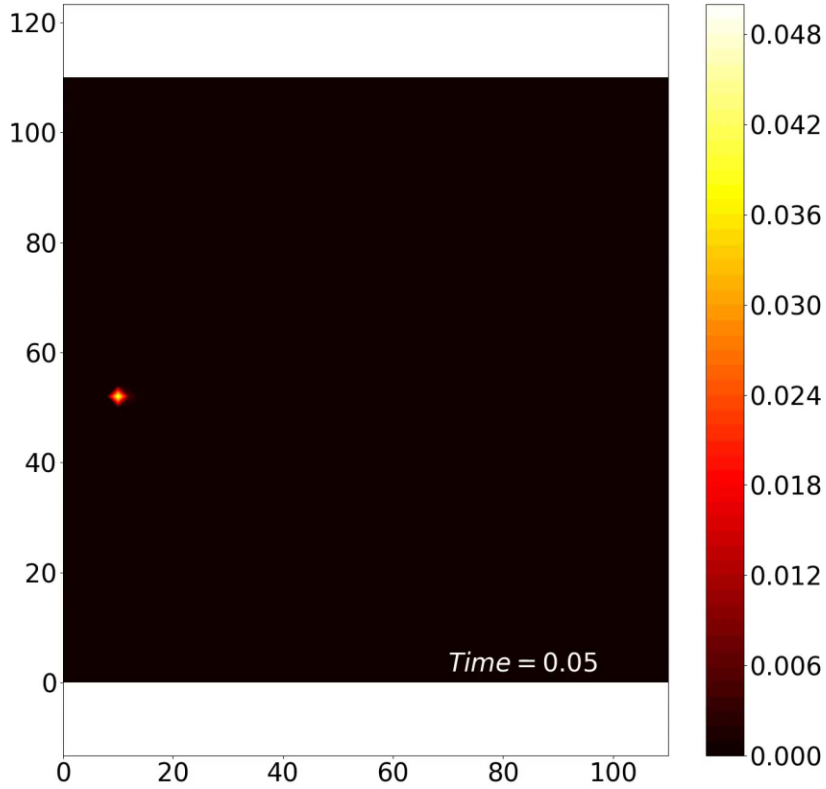
$$q(\bar{x}, t) = Q\delta(x)\delta(y)\delta(z - h)$$

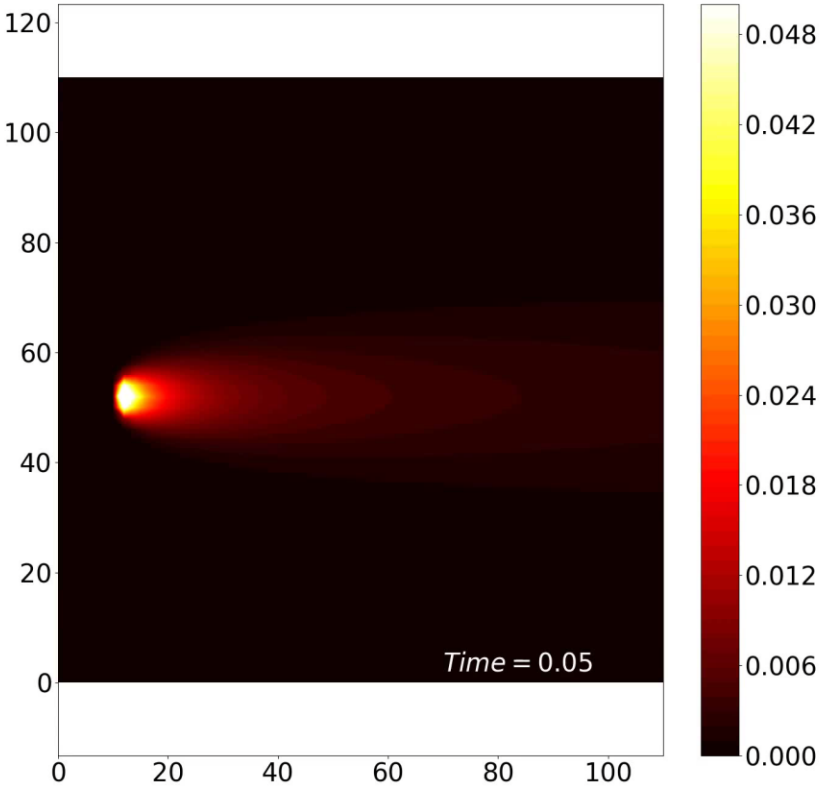
- Numerically discretized sytem to be solved:

$$\mathbf{MC}^{t+\Delta t} = \mathbf{NC}^{t} + \mathbf{q}\Delta t$$

- Here:
  - Both *M* and *N* are both $\mathrm{n} \times n$ matrices where *n* is the total number of grid points in the discretized domain.
  - *C* is the solution vector.

TE vs GD





Video : Difference between mean advected TE solution and GD solution

Errors in GD Model

- Effects to consider for a model:

1. Vertical Variability of K	2. Streamwise Diffusion	3. Wall Effect	4. Coreolis Effect

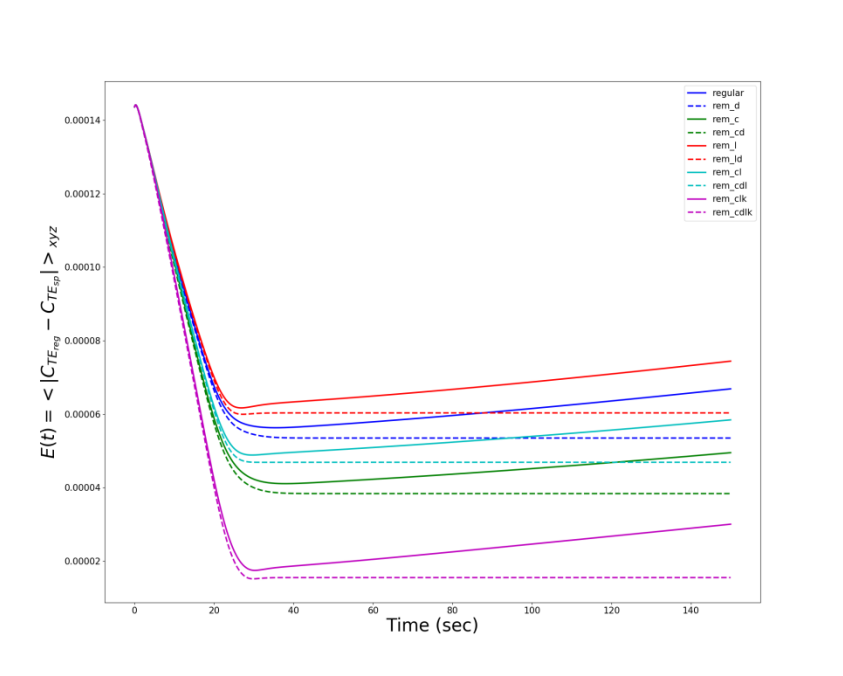


Figure a : Errors between GD and mean field advected TE with different items removed

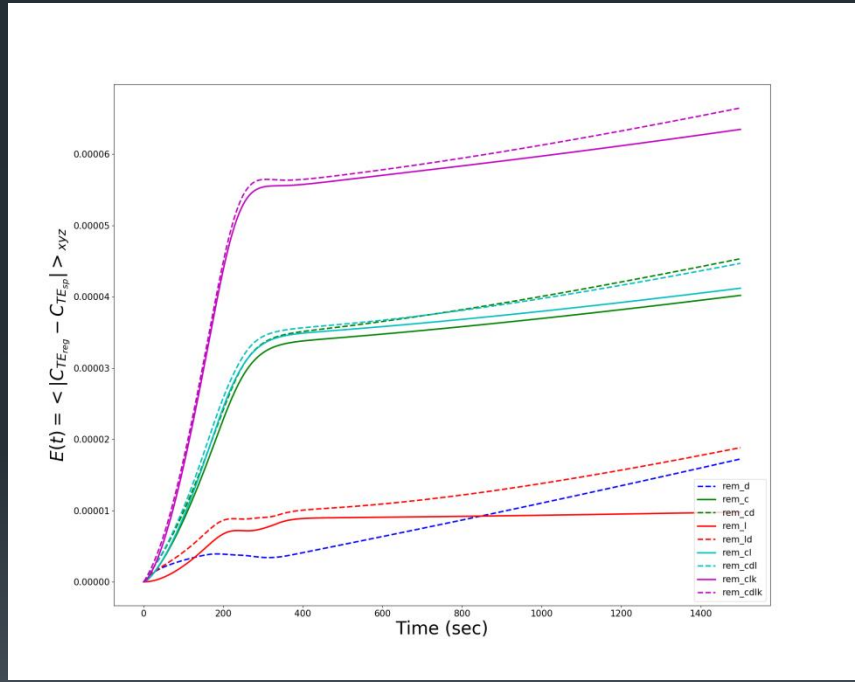


Figure b : Errors between mean advected TE with different items removed and the TE including all effects

Errors due to Diffusion

- No streamwise diffusion in GD model:

$$C(x,y,z) = \frac{Q}{2\pi\sigma_y\sigma_z u} e^{-\frac{y^2}{2\sigma_y{}^2}} [e^{-\frac{(z-h)^2}{2\sigma_z{}^2}} + e^{-\frac{(z+h)^2}{2\sigma_z{}^2}}]$$

- Diffusion timescale: $\frac{L^2}{k} \approx 7500\ sec$
- Diffusion happens slowly but consistently until diffusion timescale is reached.

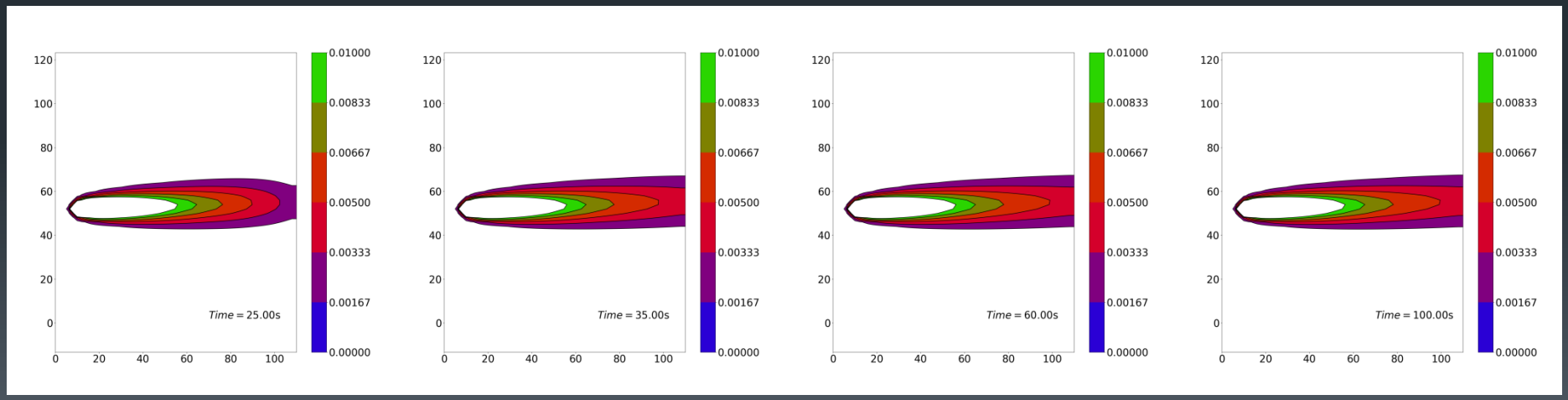


Figure : Diffusion taking place slowly
Even though advection time scale is reached

Wall Effect

- Wall effect is incorporated into GD model with the ‘z’ terms in the equation:

$$C(x,y,z) = \frac{Q}{2\pi\sigma_y\sigma_z u} e^{-\frac{y^2}{2\sigma_y{}^2}} [e^{-\frac{(z-h)^2}{2\sigma_z{}^2}} + e^{-\frac{(z+h)^2}{2\sigma_z{}^2}}]$$

- Removing wall effect from TE does not decrese the error, rather increase.
- Diffusion happens slowly but consistently until diffusion timescale is reached.
- As the vertical solution profile is non-symetric, so the wall effect is there.

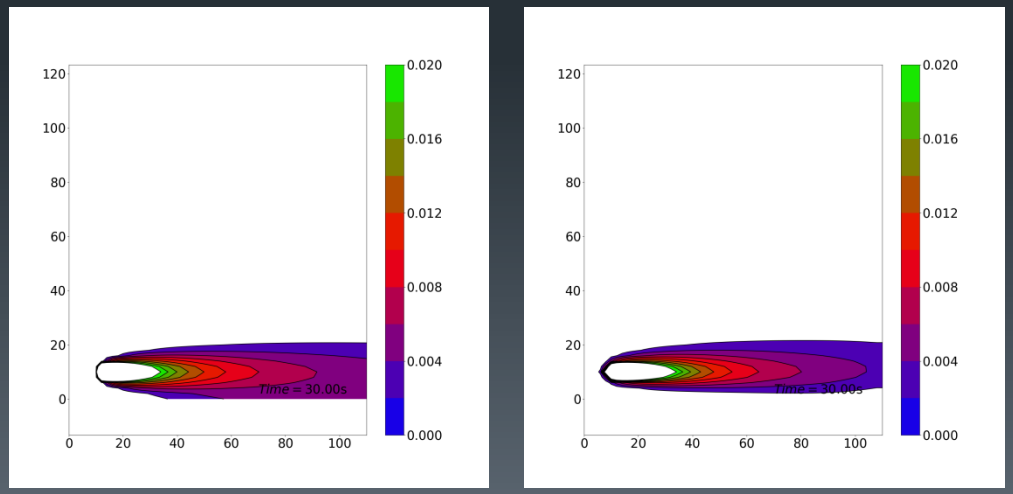


Figure : Non-symmetric vertical profile for both GD (left) and TE (right) solution

Vertical Variability of Diffusion Parameter

- In our Gaussian plume implementation we kept K constant. Thereby:

$$\sigma_y = \sigma_z = \frac{2Kx}{u}$$

- Incorporating K-variability spatially makes the solution more accurate.
- Causes biggest error in terms of margin among all other effects

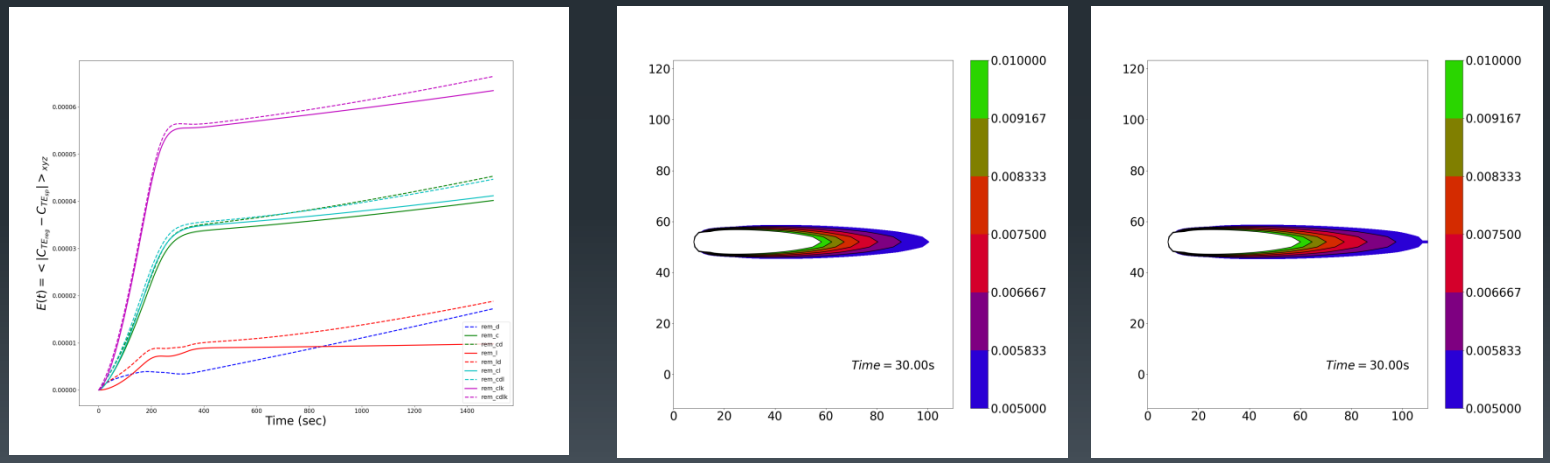


Figure : Error curves among different special cases – vs - regular TE (Left), Solution with K-variability (middle) and without it (right)

Inverse Models

- Limitation of GD based inverse model:
  - Source location and velocity both must be known apriori.
- Advantage of TE Based inverse model:
  - Both emission rate and source location can be recovered
- Limitation of TE Based inverse model:
  - Velocity field must be known apriori.
  - Prediction accuracy will fall in fully turbulent flow.

POD Based Sparse Reconstruction

- The matrix formulated in transport equation structure is used for sparse reconstruction.
- Single value decomposition (SVD) is performed on the matrix
- A measurement matrix M is built with sparsification factor 0.0001.
- Then the recovery equation is applied as following:

$$\mathbf{q} = \frac{\mathbf{M}\mathbf{C}^{t+\Delta t} - \mathbf{N}\mathbf{C}^{t}}{\Delta t}$$

- Both leak rate and source location is recovered.

Sparse Reconstruction (1.0%Data)

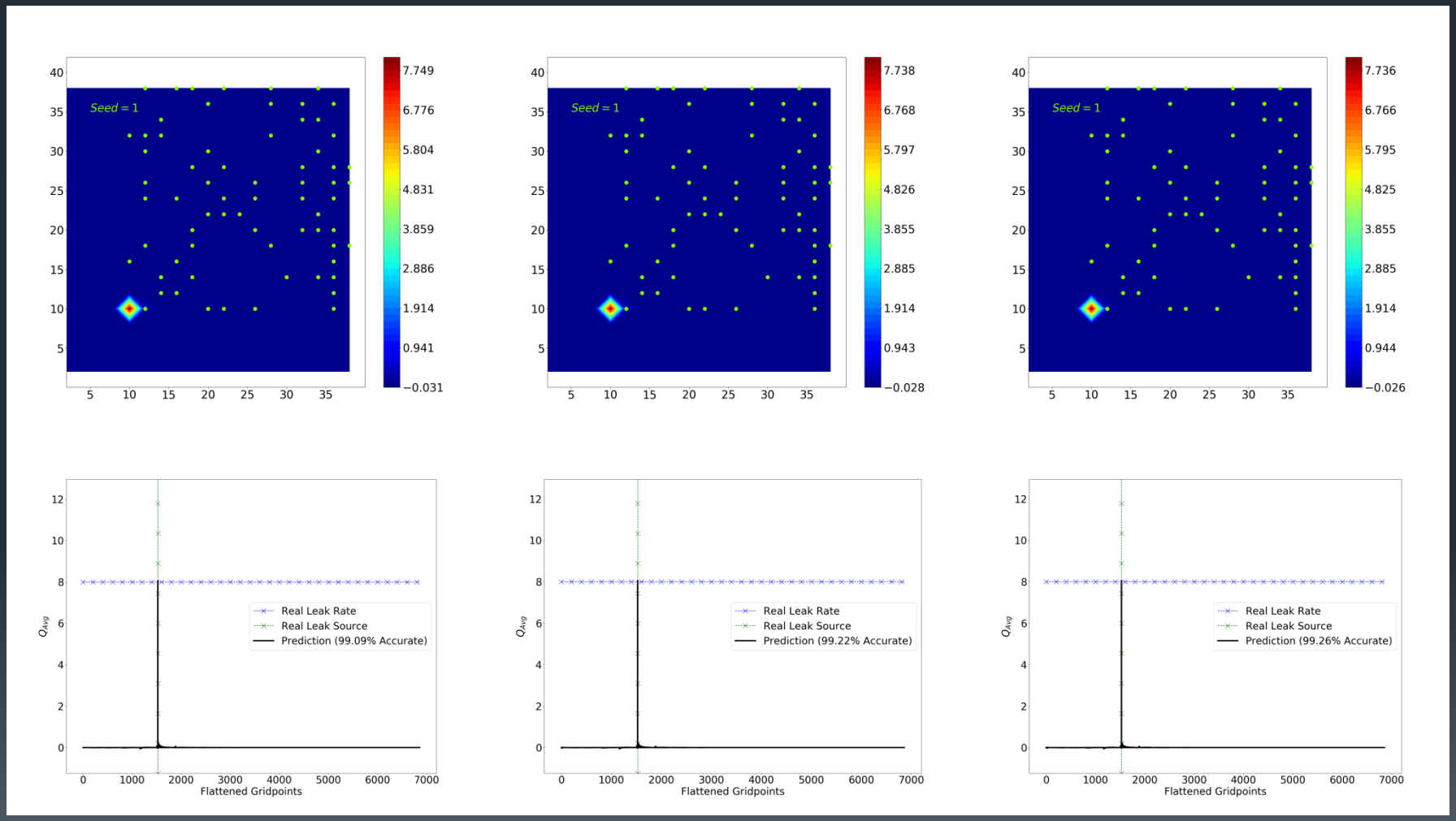


Figure: POD based sparse reconstruction using 1(left), 5(middle) and 10(right) modes (Random seed 1)

Sparse Reconstruction (0.5%Data)

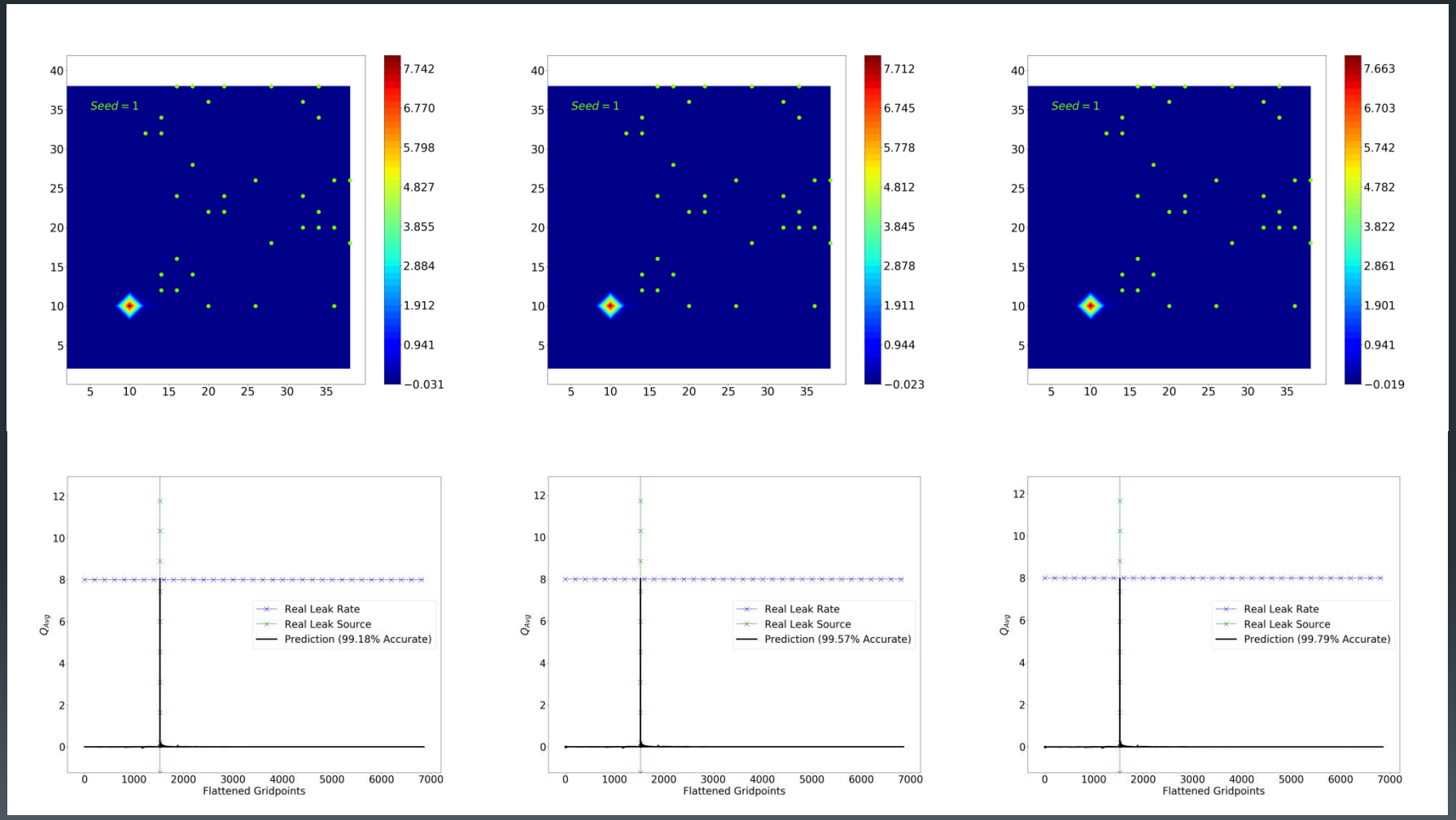


Figure: POD based sparse reconstruction using 1(left), 5(middle) and 10(right) modes (Random seed 1)

Sparse Reconstruction (0.1%Data)

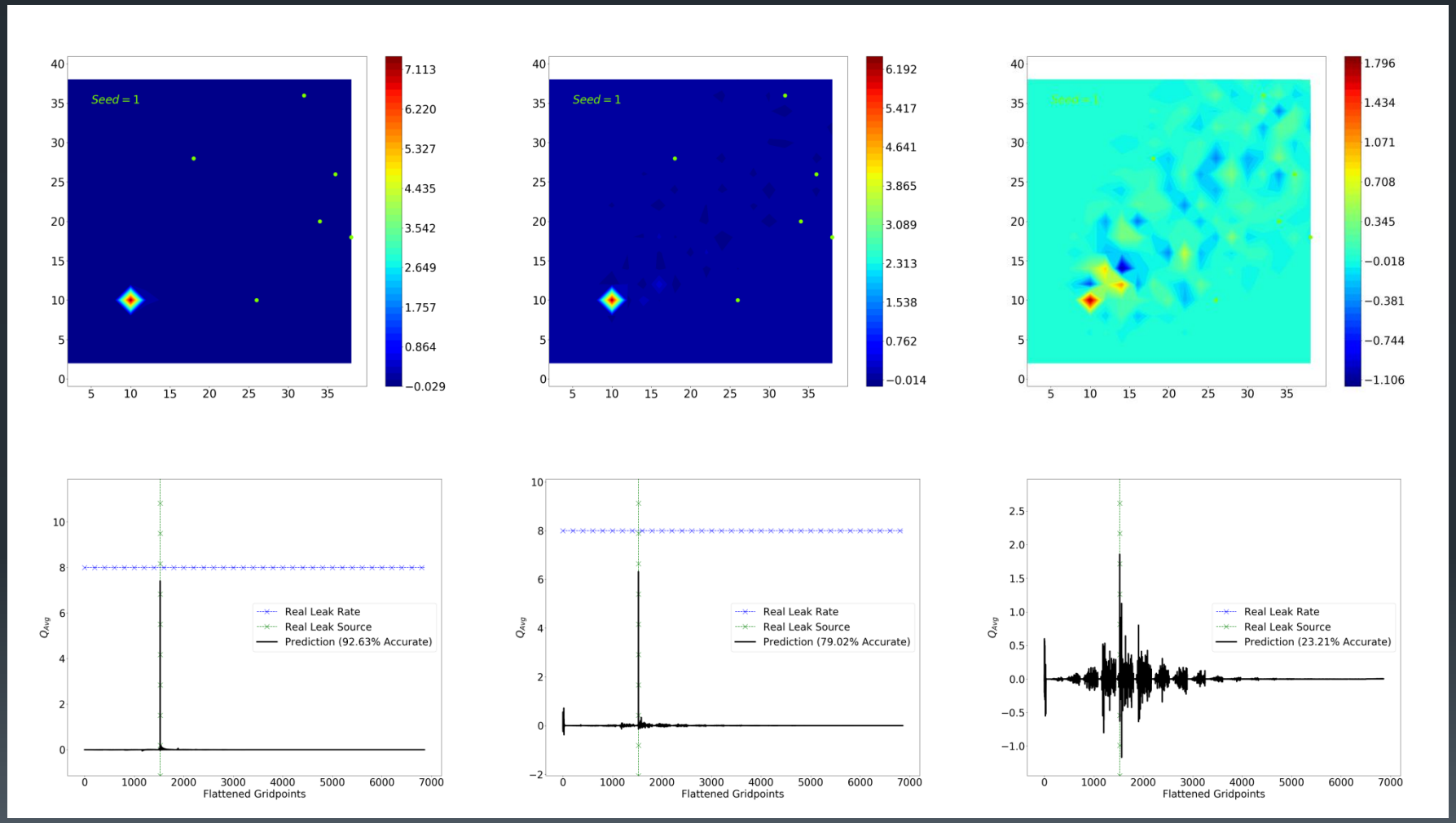


Figure: POD based sparse reconstruction using 1(left), 5(middle) and 10(right) modes (Random seed 1)

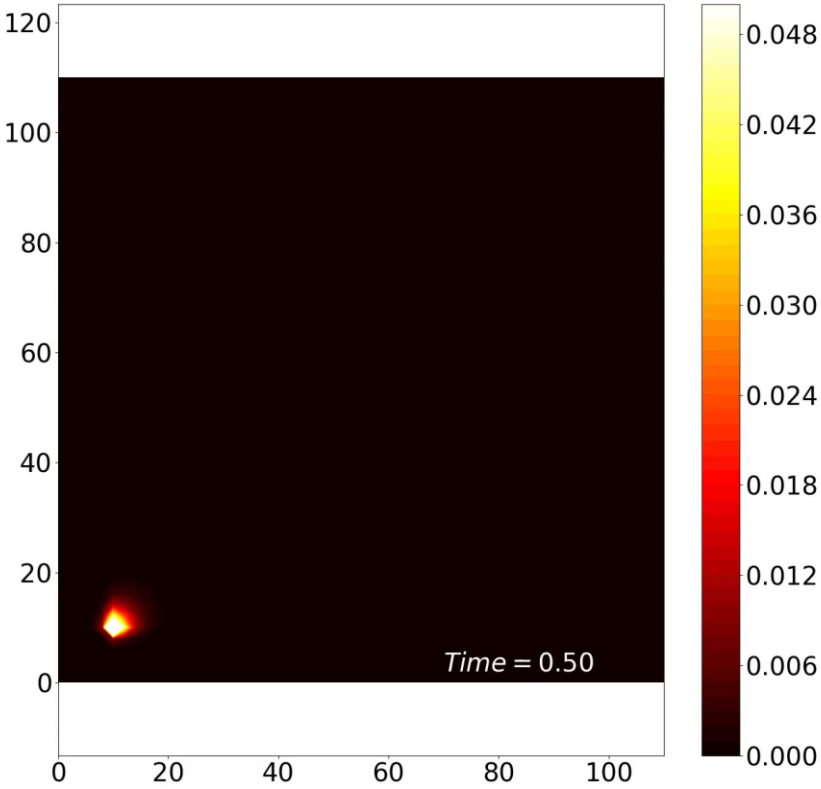
Conclusion

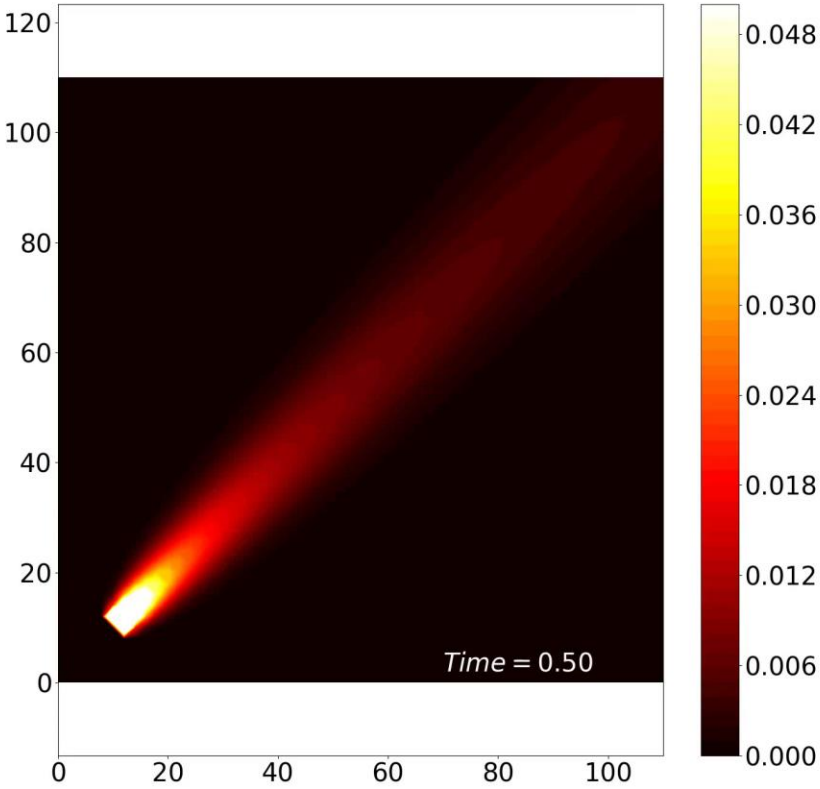
- GD is not a good model to approximate transient solution.
- K-variability has the major contribution in error between GD and TE.
- Stream wise diffusion produces relatively smaller but consistently growing error.
- Wall effect is present in the GD solution.
- Single point source location is recovered perfectly.
- Using 1 mode we can get upto 92% accurate prediction of leak rate with data as sparse as 0.1%.

Ongoing Studies & Future Prospects

- Long term goal:
  - Modeling the error between TE and GD solution.
  - Incorporating Bayesian inversion technique
- Ongoing studies:
  - Unsteady plume analysis.

TE vs GD For Unsteady Plume





Video 1: Difference between mean advected TE solution and GD solution

THANK YOU